

OLOHIMAI JULIET MICHAEL

AI / MACHINE LEARNING ENGINEERING • SYSTEMS ENGINEERING • SATELLITE & WIRELESS COMMUNICATIONS

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Portfolio: olohi184.github.io | GitHub: github.com/olohi184 | LinkedIn: linkedin.com/in/juliet-michael

PROFESSIONAL PROFILE

Engineering and AI professional with experience spanning satellite ground-station operations, mission-critical communications infrastructure, machine learning research, and deployed data-driven software. Principal Communication Engineer at Nigeria's National Space Research and Development Agency (NASRDA) and founder of WechimAI Integrated Engineering & ICT Ltd. Develops climate-aware and reliability-aware AI tools, evaluates ML models under changing operating conditions, and translates technical research into practical software applications.

CORE CAPABILITIES

AI/ML model development & evaluation • Python data analytics • Reliability and robustness assessment • Streamlit applications • Git/GitHub • Satellite and ground-station systems • RF/5G wireless networks • Systems integration • Technical research and documentation

PROFESSIONAL EXPERIENCE

Principal Communication Engineer | NASRDA, Abuja | Feb 2025–Present

- Lead and support systems-engineering work for satellite communications infrastructure, ground-station systems, technical evaluation, integration and operational reliability.
- Contribute engineering analysis, mission-support operations and data-driven problem-solving in complex communications and space-system environments.

Senior Communication Engineer | NASRDA, Abuja | Jan 2022–Feb 2025

- Supported communication systems engineering, ground-station operations and technical coordination.

Communication Engineer I | NASRDA, Abuja | Aug 2018–Dec 2021

- Worked on communications engineering and operational support in a national space-systems environment.

Founder & CEO | WechimAI Integrated Engineering & ICT Ltd | Current

- Lead an AI, automation, software and ICT consulting venture; develop demonstrations and practical tools for data analytics, workflow automation and web-based solutions.

Systems Analyst | Vivid Global Resources | Jan 2013–Jul 2018

- Professional systems analysis experience; detailed achievements to be added after verification.

Graduate Assistant / Assistant Lecturer | Covenant University | Aug 2009–Nov 2012

- Supported teaching and academic activities in information and communication engineering.

SELECTED AI AND SOFTWARE PROJECTS

ClimateNetAI / RAC-5G / Reliability-Bounded Estimation | **Founder, project originator & lead developer** | 2024–Present

- Developed research software for climate-aware 5G signal-strength prediction and evaluation using atmospheric and wireless measurements.
- Extended the work toward reliability-aware monitoring, uncertainty, degradation detection and recovery verification; RBE work investigates when a model's predictions should be treated as unreliable.
- Live application: climatenetai.streamlit.app | Code: github.com/olohi184/ClimateNetAI

N-ATLAS Reliability Lab (NARL) | **Project lead / toolkit development** | 2026

- Developed an evaluation toolkit for multilingual AI reliability, benchmarking, scoring and deployment-oriented testing.
- Worked with benchmark datasets, inference logs, automated runners and local-model evaluation workflows.

AI analytics and automation prototypes | **Developer** | Recent projects

- Built practical Python/Streamlit applications and prototypes, including an AI notes summariser, expense-tracking/data workflows, weather-data automation and inventory website.
- Portfolio demonstrations are presented as development projects, not as commissioned client engagements.

TECHNICAL SKILLS

Programming and data: Python, Pandas, NumPy, MATLAB, data cleaning, exploratory analysis, visualization.

Machine learning: scikit-learn, XGBoost, TensorFlow/Keras, PyTorch; Linear Regression, Decision Trees, Random Forest, LSTM, GRU; model comparison and time-series evaluation.

Applications and workflow: Streamlit, Git, GitHub, Google Colab, HTML/CSS, software prototyping and deployment.

Engineering domains: satellite communications, ground-station systems, RF/wireless networks, 5G signal metrics, environmental sensing, cybersecurity and privacy.

RESEARCH AND ENGINEERING CONTRIBUTIONS

- Designed comparative ML/DL studies using longitudinal 5G and atmospheric measurements, including seasonal and temporal evaluation.
- Investigated distribution shift and limitations of conventional holdout validation, emphasizing reliability under real operating conditions.
- Presented research on generalization for climate-aware 5G prediction at WiML @ NeurIPS 2026 (poster acceptance).

EDUCATION

M.Eng., Information and Communication Engineering | Covenant University, Nigeria | 2012

B.Eng., Information and Communication Engineering | Covenant University, Nigeria | 2008

SELECTED PUBLICATIONS

- Iruemi, O. J., et al. (2025). “Machine Learning-Based Analysis of Weather Influence on 5G Network Signal Propagation in North-Central Nigeria.” Journal of Applied Bioanalysis.
- Iruemi, O. J., et al. (2024). “Assessment of Seasonal Variation of 5G Signal Strength and Signal Speed over Abuja, North-Central Nigeria.” Journal of Electrical Systems.
- Ale, F., Ayegba, A., Momoh, J., Ikpaye, D. I., & Iruemi, O. J. (2025). “Strengthening Data Privacy and Protection in Nigeria: Recommendations for a Comprehensive Approach.” 8th URSI-NG Annual Conference. DOI: 10.2991/978-94-6463-644-4_4.
- Ayegba, A., Iruemi, O. J., et al. (2025). “Enhanced Security Features of 6G Network: A Leap Beyond 5G Network.” American Journal of Technology Advancement.

RESEARCH RECOGNITION AND ENGAGEMENT

- Poster presenter, Women in Machine Learning (WiML) @ NeurIPS 2026 — “When Simpler Models Win: Evaluating Machine Learning Generalization for Climate-Aware 5G Signal Prediction.”
- Reviewer, Machine Learning for Physical Sciences (ML4PS), 2026.
- ICEGOV 2025 Best Paper Award (as listed in supplied academic CV; award attribution can be expanded).
- Participant, ACM AI Leadership Summit 2026, Atlanta, Georgia, USA (as listed in supplied academic CV).

PROFESSIONAL MEMBERSHIPS

ACM • IEEE • IEEE Women in Engineering • INCOSE • COREN (Registered Engineer) • Nigerian Society of Engineers • Nigeria Computer Society

LEADERSHIP AND OUTREACH

- Founder and technology entrepreneur through WechimAI Integrated Engineering & ICT Ltd, focusing on practical AI and digital solutions.
- STEM and technology outreach supporting girls and women in computing, engineering and AI skills development.

SELECTED LINKS

Professional portfolio: <https://olohi184.github.io/>

GitHub: <https://github.com/olohi184>

ClimateNetAI live app: <https://climatenetai.streamlit.app>

Google Scholar: <https://scholar.google.com/citations?user=4Ma0pYwAAAAJ>

LinkedIn: <https://www.linkedin.com/in/juliet-michael/>

Note: Earlier scholarly publications appear under the author name Iruemi, O. J.